

Machine Learning

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This document consists of personal, independent, and unofficial lecture notes and summaries covering topics from the **Machine Learning** course for the **Computer Engineering** degree program. This material is created solely to support individual study and is shared free of charge for peer collaboration. Sale or commercial distribution is strictly prohibited.

Introduction

Note: The core course notes officially begin in Chapter 1 (the preceding sections serve as introductory background—feel free to skip them if you wish).

Machine Learning provides frameworks and tools to solve decision-making tasks under limited information provided by measured data. The fundamental components of learning are:

1. **Input** $x \in \mathcal{X}$: a vector of p features from which we make predictions.
2. **Output** $y \in \mathcal{Y}$: the target value we want to predict.

Our goal is to learn a target function:

$$f: \mathcal{X} \rightarrow \mathcal{Y}$$

The data consists of a set of examples:

$$S = \{(x_1, y_1), (x_2, y_2), \dots, (x_N, y_N)\}$$

which forms the **training set** (typically assumed to be an i.i.d. sample drawn from a distribution $\mathcal{D}$).

After training on this dataset, our algorithm outputs a **hypothesis function** $h: \mathcal{X} \rightarrow \mathcal{Y}$ chosen from a hypothesis class $\mathcal{H}$ ($h \in \mathcal{H}$) that approximates the true function f as closely as possible.

Machine Learning problems can be categorized into several paradigms:

- **Classification:** Supervised learning where $\mathcal{Y}$ is discrete.
- **Regression:** Supervised learning where $\mathcal{Y}$ is continuous.
- **Clustering:** Unsupervised learning where $\mathcal{Y}$ is discrete.
- **Dimensionality Reduction:** Unsupervised learning where $\mathcal{Y}$ is continuous.

Before diving deeper into Machine Learning topics, we must review the three core pillars that underpin the field: Probability and Statistics, Linear Algebra and Optimization, and Computation (algorithms and data structures).

Statistics

Estimating the Mean of a Gaussian Random Variable

Suppose we observe a dataset of independent and identically distributed (i.i.d.) random variables, and we assume $X_i \sim \mathcal{N}(\theta, \sigma^2)$. Our goal is to estimate the unknown mean θ using the realizations $x_1, \dots, x_m$. To do this, we need to:

1. Find a point estimator $\hat{\theta} = g(X_1, \dots, X_m)$.
2. Determine how close this point estimator is to the true mean θ .
3. Decide whether it is plausible that θ equals a specific value (for example, whether the data supports the claim that $\theta = 0$).

Probability vs. Likelihood. **Probability** answers the question: "If the parameter θ is known, how likely are we to observe the data x ?"

Likelihood inverts this perspective: "Given that we have observed data $x = [x_1, \dots, x_m]$ but do not know the true parameter θ , which parameter value makes our observed data most plausible?"

The **likelihood function** $p_X(x; \theta)$ is the joint probability density of the sample vector x parameterized by the unknown parameter θ . For a fixed observation x , it evaluates how the plausibility of the observed data varies as a function of θ .

Because products of probability densities turn into convenient sums under monotonic transformations, we define the **log-likelihood function**:

$$L(\theta) := \log p_X(x; \theta)$$

Example: Gaussian Mean Estimation. For $X_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\theta, \sigma^2)$, the density for a single sample is:

$$p_{X_i}(x_i; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \theta)^2}{2\sigma^2}\right)$$

The log-likelihood of the full sample is therefore:

$$L(\theta) := \log p_X(x; \theta) = \log\left(\prod_{i=1}^m p_{X_i}(x_i; \theta)\right) = -\frac{m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^m (x_i - \theta)^2$$

(Verifying these algebraic steps is left as a simple exercise).

The **Maximum Likelihood Estimator (MLE)** of θ is defined as:

$$\hat{\theta} := \arg \max_{\theta} L(\theta)$$

In words, $\hat{\theta}$ is the parameter value that maximizes the log-likelihood function $L(\theta)$. Searching for this maximum is well motivated because, as the sample size m grows large, the maximizer converges to the true population mean.

Solving the first-order condition $\frac{d}{d\theta} L(\theta) = 0$ yields:

$$\hat{\theta} = \frac{1}{m} \sum_{i=1}^m x_i$$

which is simply the sample mean. Note that if we observe only a single measurement x_1 , the value of $\hat{\theta}$ that maximizes the likelihood is simply x_1 itself.

Confidence Intervals. A point estimate alone is not sufficient because the estimator itself is a random variable. If we repeat the experiment tomorrow with a fresh batch of samples, we will obtain a slightly different value for $\hat{\theta}$. A single number does not tell us how close our estimate is to the true θ .

Instead of a single point, we construct a data-dependent set or interval $I(X)$ centered around $\hat{\theta}$ that is likely to contain the true parameter θ .

Important conceptual distinction: The interval $I(X)$ is a **random set** (since it depends on the random sample X), whereas the parameter θ is an **unknown, fixed constant**.

We say that $I(X)$ is a **confidence interval** for θ at confidence level $1 - \alpha$ if:

$$\Pr[\theta \in I(X)] \geq 1 - \alpha$$

In frequentist terms, this means that across repeated independent experiments drawing samples $x \sim p_X(x; \theta)$, approximately $(1 - \alpha) \times 100\%$ of the constructed intervals will contain the true, unknown parameter θ .

Conceptually, the interval can be expressed as:

$$I(X) = \text{Center} \pm \text{Margin of Error}$$

Ideally, we want $I(X)$ to be narrow (providing high precision) while having a small α (providing high confidence).

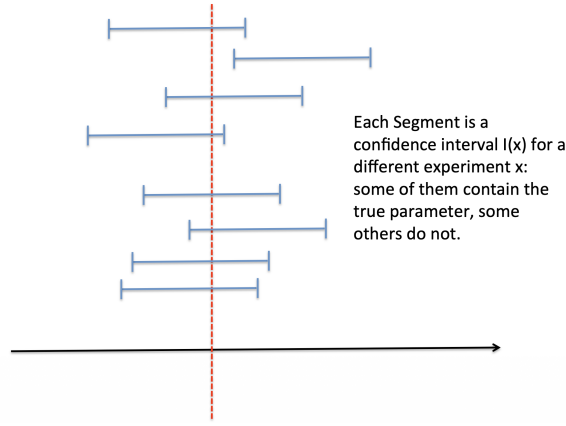


Figure 1: Example of confidence intervals across repeated samples.

Since $\hat{\theta} = \frac{1}{m} \sum_{i=1}^m X_i \sim \mathcal{N}\left(\theta, \frac{\sigma^2}{m}\right)$, one can readily show that for a given confidence level $1 - \alpha$, the narrowest confidence interval for θ is symmetric and centered at the sample mean:

$$I(X) = \left[\hat{\theta} - z_{\alpha/2} \frac{\sigma}{\sqrt{m}}, \hat{\theta} + z_{\alpha/2} \frac{\sigma}{\sqrt{m}} \right]$$

Hypothesis Testing. Often, rather than just estimating θ , we want to test a hypothesis about its value (the **null hypothesis** H_0). For instance, we may wish to test whether $\theta = \theta_0$ for some fixed, known value θ_0 .

Before observing the data, we partition the sample space into two regions:

- **Acceptance Region:** Outcomes that are plausible and expected under H_0 .
- **Rejection Region $\mathcal{R}(H_0)$:** Extreme or unlikely outcomes under H_0 that lead us to reject H_0 .

Given an observed sample $x \sim p_X(x; \theta)$, our decision rule is:

$$x \in \mathcal{R}(H_0) \implies \text{Reject } H_0$$

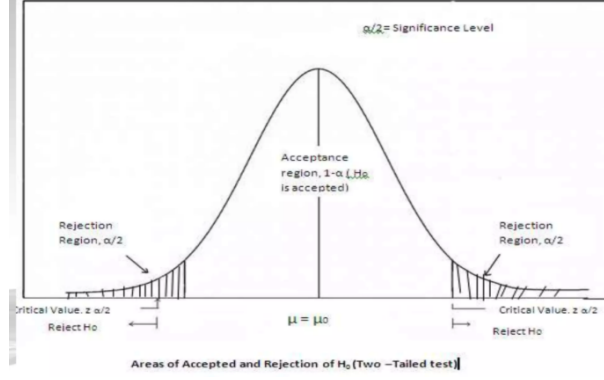


Figure 2: If the true distribution has mean μ_0 , sample realizations falling within the rejection tails are considered too far from the expected value to be consistent with H_0 .

This decision rule can result in two types of errors:

1. **Type I Error:** Rejecting H_0 when H_0 is true.
2. **Type II Error:** Failing to reject H_0 when H_0 is false.

Their respective probabilities are denoted by:

$$\Pr[x \in \mathcal{R}(H_0) \mid H_0 \text{ is true}] = \alpha \quad (\text{Significance Level})$$

$$\Pr[x \notin \mathcal{R}(H_0) \mid H_0 \text{ is false}] = \beta$$

Evaluating the Type I error probability is typically more straightforward because the null distribution is fully specified. Note that fixing the significance level α does not uniquely determine the rejection region; it only specifies the total probability mass (e.g., 5%) allocated to the rejection region under the null hypothesis. That probability mass could be placed entirely in the upper tail, entirely in the lower tail, or split equally between both tails (2.5% each).

Gaussian Test Example. Returning to our Gaussian mean example (assuming for simplicity that $\sigma^2 = 1$), suppose we test the null hypothesis:

$$H_0: \theta = \theta_0 = 0$$

The Neyman-Pearson lemma ensures that the most powerful symmetric two-sided rejection region for a fixed Type I error rate α has the form:

$$\mathcal{R}(H_0) = \left\{ (x_1, \dots, x_m) : |\hat{\theta} - \theta_0| = \left| \frac{1}{m} \sum_{i=1}^m (x_i - \theta_0) \right| > \frac{z_{\alpha/2}}{\sqrt{m}} \right\}$$

Setting $\theta_0 = 0$, this simplifies to:

$$\mathcal{R}(H_0) = \left\{ (x_1, \dots, x_m) : |\hat{\theta}| = \left| \frac{1}{m} \sum_{i=1}^m x_i \right| > \frac{z_{\alpha/2}}{\sqrt{m}} \right\}$$

Since under H_0 we have $\hat{\theta} \sim \mathcal{N}\left(0, \frac{1}{m}\right)$, the probability of committing a Type I error directly matches our target significance level α :

$$\Pr \left[|\hat{\theta}| > \frac{z_{\alpha/2}}{\sqrt{m}} \mid H_0 \right] = \alpha$$

The p -value. Instead of comparing the test statistic directly to a predetermined critical threshold, we can summarize the test using the p -value.

Given the observed sample mean $\bar{x} = \frac{1}{m} \sum_{i=1}^m x_i$, the p -value is the probability of observing a test statistic at least as extreme as the one computed from the data, assuming H_0 is true:

$$p := \Pr \left[|Z| > \left| \frac{1}{m} \sum_{i=1}^m x_i \right| \middle| H_0 \right], \quad \text{where } Z \sim \mathcal{N} \left(0, \frac{1}{m} \right)$$

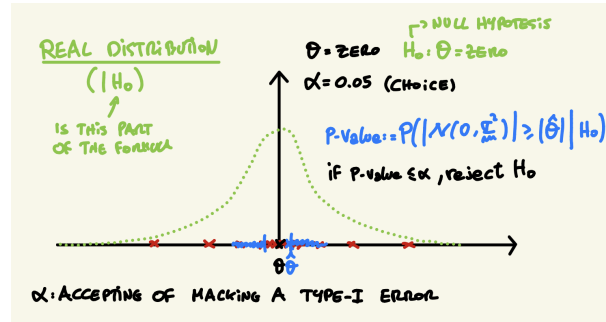


Figure 3: Visual representation of the two-tailed p -value area under the null distribution.

Duality Between Confidence Intervals and Hypothesis Tests. There is a direct mathematical duality connecting these two concepts: rejecting the null hypothesis $H_0: \theta = \theta_0$ at significance level α is completely equivalent to checking whether the hypothesized value θ_0 falls outside the corresponding $(1 - \alpha)$ confidence interval $I(X)$.

Linear Regression with Known Variance σ^2

To be continued...

Chapter 1

In supervised learning, our goal is to infer a rule from observed data. We start by formally defining the spaces:

- **Domain Set ($\mathcal{X}$):** The space of all possible inputs (features).
- **Label Set ($\mathcal{Y}$):** The space of all possible outputs (targets). Depending on the problem, $\mathcal{Y}$ can be:
 - *Discrete*: in classification problems (e.g., recognizing handwritten digits from an image).
 - *Continuous*: in regression problems.

The learner does not see the entire world, but only a finite set of labeled examples called the **Training Set** (or sample set):

$$S := \{(x_i, y_i)\}_{i=1}^m$$

where m denotes the sample size.

How does the environment generate data?

To analyze learning mathematically, we must make assumptions about how the training set S is produced:

1. **Simple Assumption (Deterministic Labels):** Each instance x_i is sampled independently and identically distributed (i.i.d.) from a probability distribution $\mathcal{D}_x$, and there exists an unknown deterministic ground-truth function $f(x)$ that maps each x_i to its exact label:

$$x_i \sim \mathcal{D}_x \quad \text{and} \quad \exists f(x) \text{ such that } y_i = f(x_i)$$

2. **Realistic Assumption (Probabilistic Labels):** In the real world, identical inputs may have different labels due to noise or missing information. Thus, the output y is also governed by a distribution conditioned on x :

$$y \mid x \sim \mathcal{D}_{y|x}$$

Key Insight: The *Realistic* setting is the general framework. The *Simple* setting is just a special case where $\mathcal{D}_{y|x}$ is completely deterministic—meaning all the probability mass is concentrated at the single value $f(x)$ (conceptually like a Dirac delta δ function).

2. The Learning Algorithm and Loss Functions

Under the deterministic assumption, our concrete goal is: **learn $f(x)$ from empirical measurements to predict y for unseen inputs x .**

A **Learning Algorithm** A takes the training set S as input and outputs a hypothesis function h from a hypothesis class $\mathcal{H}$:

$$A(S) \rightarrow h \in \mathcal{H}$$

Here, h acts as our best candidate approximation of the true function f .

Quantifying Error: The Loss Function

To measure how well a chosen hypothesis h predicts y , we define a **Loss Function** $\ell(h, x)$.

Important distinction: The loss function evaluates the discrepancy on a **single sample**!

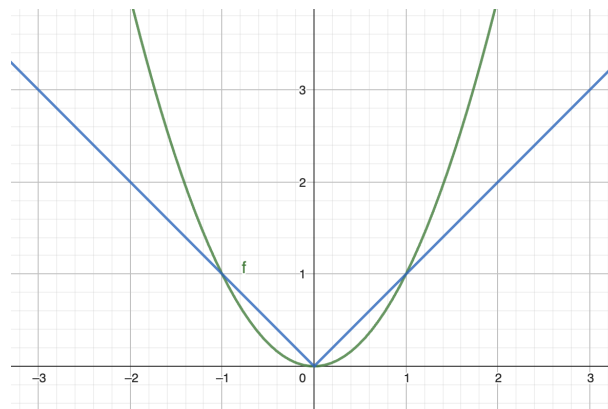
Example 1: Classification

The most standard choice in classification is the **0–1 loss function**:

$$y \in \{0, 1\}, \quad \ell_{0-1}(h, x) = \begin{cases} 0 & \text{if } h(x) = y \quad (\text{correct classification}) \\ 1 & \text{if } h(x) \neq y \quad (\text{misclassification}) \end{cases} \quad (1)$$

Example 2: Quadratic Loss vs. Absolute Loss

The mathematical choice of the loss function heavily affects how the algorithm treats mistakes:



Consider comparing:

$$\text{Squared Error: } \ell(h, x) = (y - h(x))^2 \quad (y \in \{0, 1\} \text{ or } \mathbb{R})$$

vs.

$$\text{Absolute Error (Modulus): } \ell(h, x) = |y - h(x)| \quad (y \in \mathbb{R})$$

Takeaway for students:

- The **quadratic loss** penalizes larger errors much more heavily (growing quadratically), making it sensitive to outliers.
- The **modulus loss** increases linearly with the error magnitude, maintaining constant sensitivity regardless of scale.

3. From Single Samples to Global Error: True Risk vs. Empirical Risk

Evaluating the error on a single instance (x, y) is not enough: we want to know how well h performs on average across all possible instances in the environment. This average is the **True Risk** (or Generalization Error):

$$L_{\mathcal{D}}(h) := \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(h, x)]$$

$L_{\mathcal{D}}(h)$ represents the true, expected error of h over the entire distribution $\mathcal{D}$. Therefore, the ideal learning algorithm would pick the hypothesis that minimizes this quantity:

$$A(S) = \arg \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h)$$

The Big Problem

To compute $L_{\mathcal{D}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(h, (x, y))]$, we would need complete knowledge of the underlying distribution $\mathcal{D}$. But in real-world machine learning, $\mathcal{D}$ is **unknown**!

The Solution: Empirical Risk

Since we cannot compute the theoretical expectation directly, we compute the empirical sample mean over our available training data S .

By the **Law of Large Numbers (LLN)**, as the sample size grows to infinity ($m \rightarrow +\infty$), the sample mean $\hat{\theta}$ converges to the true population mean θ :

$$\hat{\theta} \approx \theta$$

This justifies calculating the **Empirical Risk** $L_S(h)$ directly from our dataset:

$$L_S(h) := \frac{1}{m} \sum_{i=1}^m \ell(h, (x_i, y_i))$$

—

4. The Ultimate Goal: PAC Learning

Our ultimate goal is **not** just to minimize $L_S(h)$ on the training data (doing so blindly can lead to *overfitting*, which we will analyze later), but to guarantee that the **True Risk** $L_{\mathcal{D}}(h)$ is as **small as possible**.

Because the training set S is drawn randomly, there is always a tiny probability of drawing an unrepresentative sample. Therefore, we look for guarantees of the form:

$$\mathbb{P}[\text{True Risk } L_{\mathcal{D}}(A(S)) \text{ is small}] \text{ is large}$$

In mathematical terms, we want:

$$\mathbb{P}[L_{\mathcal{D}}(A(S)) \leq \epsilon] \geq 1 - \delta$$

This core framework is known as **PAC Learning** (*Probably Approximately Correct Learning*):

- **Approximately Correct:** The error $L_{\mathcal{D}}(h)$ is bounded by a small threshold ϵ .
- **Probably:** This condition holds with high confidence $1 - \delta$.

To be continued